

Fig. 2.20 Showing repulsions.

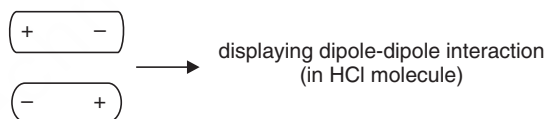
Highlight:

NH_3 and H_2O molecules are examples of distorted molecules due to the presence of lone pairs in the hybrid orbitals and lp-bp, lp-lp repulsions.

Intermolecular forces:

In a crystalline solid, the particles act as structural units, ions of molecules are arranged in a symmetrical way. So melting of such solids occurs when the highly ordered arrangement of particle in the crystalline lattice changes to the more random arrangements (characteristic of a liquid). In non-ionic compound (covalent compound) where the atoms are held by covalent bonds, the intermolecular forces are very weak. So energy requirement for melting is less compared to ionic compounds. There are two kinds of intermolecular forces:

- (a) dipole-dipole interaction,
- (b) van der Waals forces.



As a result of dipole-dipole interaction, polar molecules are held to each other more strongly than non-polar molecules (purely covalent molecules). This difference in strength of intermolecular forces is reflected in their physical properties.

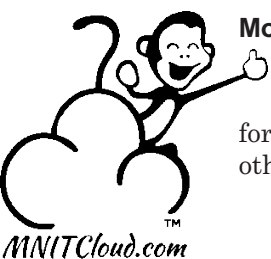
There must be some forces between the molecules of a non-polar compound. Since, they can solidify. Such attraction is known as *van der Waals force of attraction*. These van der Waals forces have very short range. They act only between the portions of different molecules that are in close contact, *i.e.*, van der Waals forces are weak intermolecular forces which are responsible for weakest attractions between nonpolar molecules.

Every atom has an effective 'size'. This is called van der Waals radius. When two non-bonded atoms come closer, the attraction between them steadily increases and reaches a maximum when they are just "touching", when the distance between the nuclei is equal to the sum of the van der Waals radii, if the atoms are forced to come still closer then *van der Waals attraction* will be replaced by *van der Waals repulsion*. Thus, we can say, non-bonded atoms strongly resist crowding. Thus, both attractive and repulsive van der Waals forces are important for our understanding of molecular structure.

Molecular orbital theory [linear combination of atomic orbitals, LCAO]

Homonuclear diatomic molecule:

Let, two H-atoms designated by H_A and H_B containing two 1s-atomic orbitals overlap to form M.O. Then expression for two molecular orbitals (M.O.) will be one bonding (ϕ_B) and the other antibonding (ϕ_A):



$$\phi_B = \psi_A(1s) + \psi_B(1s)$$

$$\phi_A = \psi_A(1s) - \psi_B(1s)$$

(i) The two conditions can be represented pictorially (Fig. 2.21)

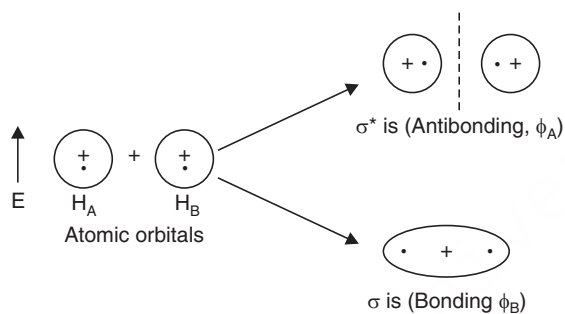


Fig. 2.21 Bonding and antibonding orbitals.

Linear combination of 1s-atomic orbitals (σ -symmetry) of two H-atoms to give bonding and antibonding molecular orbitals (σ -symmetry).

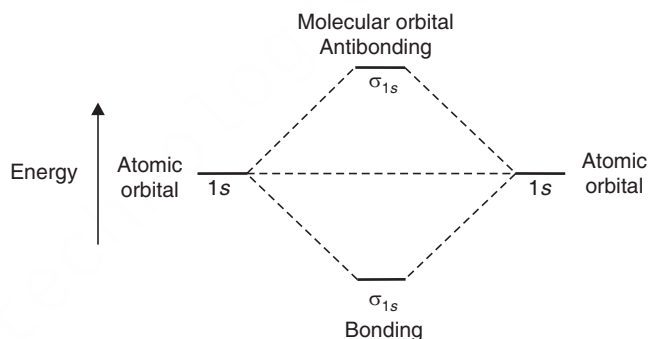


Fig. 2.22 Representation of M.O. diagram.

Energy level diagram (Fig. 2.22) illustrating the formation and relative energies of σ_{1s} (bonding) and σ_{1s}^* (antibonding) molecular orbitals resulting from the combination of two 1s-atomic orbitals on two H-atoms in H_2 molecule.

(ii) Molecular orbitals from p atomic orbitals:

(a) End-end overlap

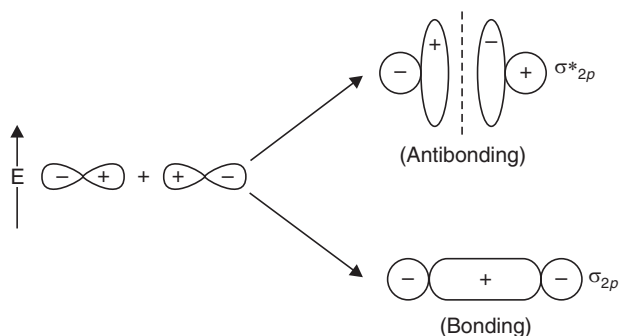
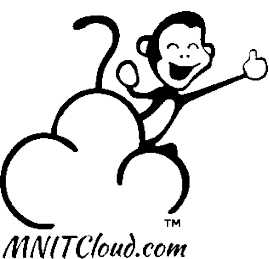


Fig. 2.23 Linear combination of two $2p_x$ orbitals to form σ_{2p} (bonding) and σ_{2p}^* (antibonding) molecular orbitals (σ -symmetry).



(b) Side-by-side overlap (Fig. 2.24):

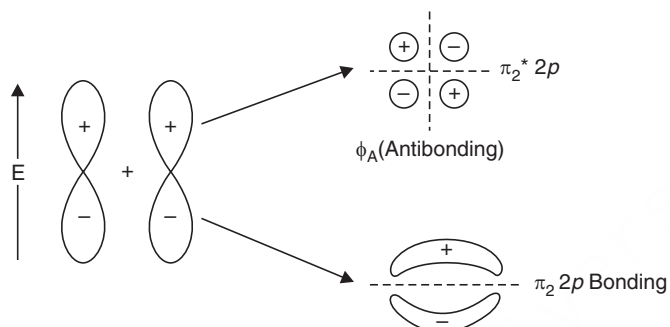


Fig. 2.24 π -Bonding.

Linear combination of two $2p_z$ orbitals to form $\pi_z 2p$ (bonding) and $\pi_z^* 2p^*$ (antibonding) molecular orbitals (π -symmetry).

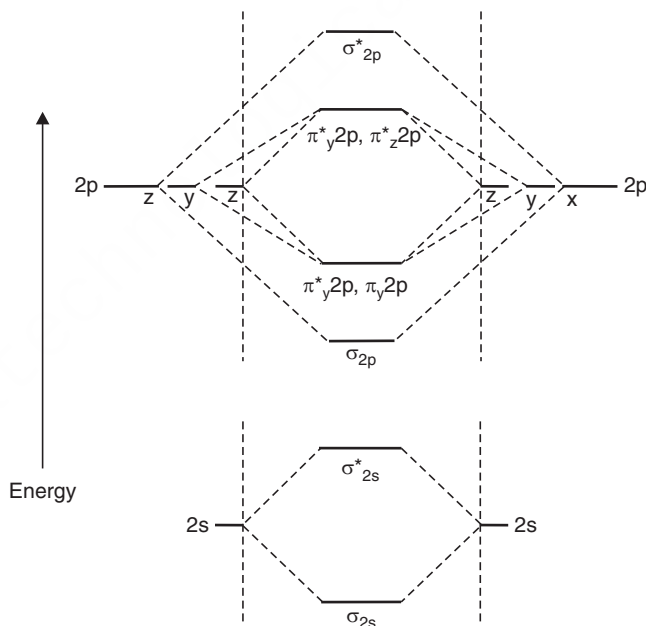


Fig. 2.25 Order of energy of molecular orbitals (M.O.).

(Energy level diagram of molecular orbitals for a homonuclear diatomic molecule)

So, the order of energy for molecular orbital is:

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2p} < \frac{\pi_y 2p}{\pi_z 2p} < \frac{\pi_y^* 2p}{\pi_z^* 2p} < \sigma_{2p}^*$$

Bond-Order (B.O.):

Bond-order (B.O.) of a molecule is calculated as follows:

$$\text{B.O.} \Rightarrow \frac{\text{N.B.} - \text{N.A.}}{2}$$

N.B. $\Rightarrow$ no. of electrons in bonding molecular orbitals.

N.A. $\Rightarrow$ no. of electrons in antibonding molecular orbitals.

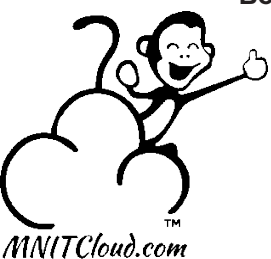


Table 2.4

B.O.	Significance
0	molecule is unstable and does not exist.
-ve	unstable and does not exist.
+ve	*stable and exists.

*Greater the value of B.O. greater is the stability of the molecule.

• *Rules for filling up of molecular orbitals with electrons:*

- The molecular orbitals are filled in the increasing order of their energies. Thus molecular orbital possessing the lowest energy will be filled first and that possessing the highest energy will be filled up in the last.
- The maximum number of electrons that can fill a molecular orbital is two.
- According to Hund's rule of maximum multiplicity, if there are two molecular orbitals having the same energy (*i.e.*, degenerate molecular orbitals), the electrons will first fill them singly and when both of them have been filled singly, pairing of electrons will occur. Thus $\pi_y 2p$ and $\pi_z 2p$ bonding molecular orbitals which are degenerate are first singly filled and pairing takes place only when more electrons are to be accommodated. In a similar fashion $\pi_y^* 2p$ and $\pi_z^* 2p$ antibonding degenerate molecular orbitals are filled just like degenerate bonding orbitals.

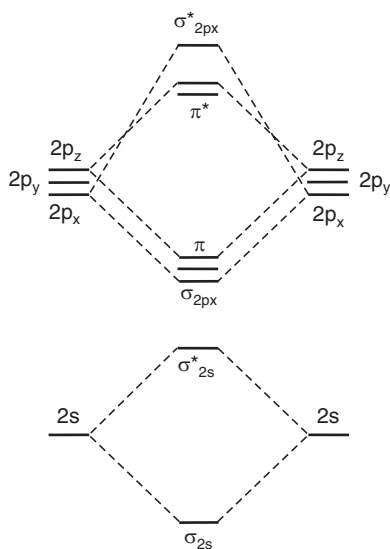


Fig. 2.26 Molecular orbital diagram from 2s and 2p atomic orbitals.

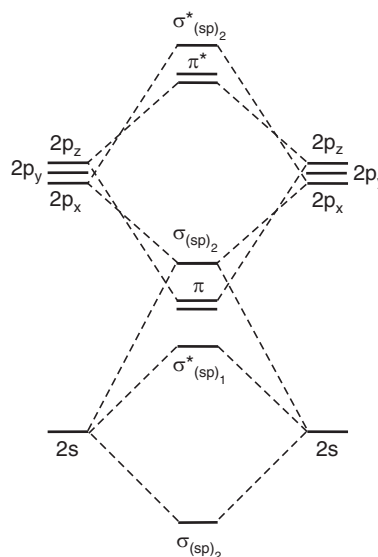
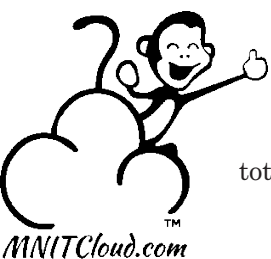


Fig. 2.27 Molecular orbital diagram allowing s-p interaction.

• *Energy level diagram for N_2 molecule:*

The electronic configuration of N-atom is $1s^2 2s^2 2p^3$. In N_2 molecule there will be all total 14 electrons of which 4 will be in the K shells which is denoted by KK.



So, the electronic configuration of N_2 molecule will be

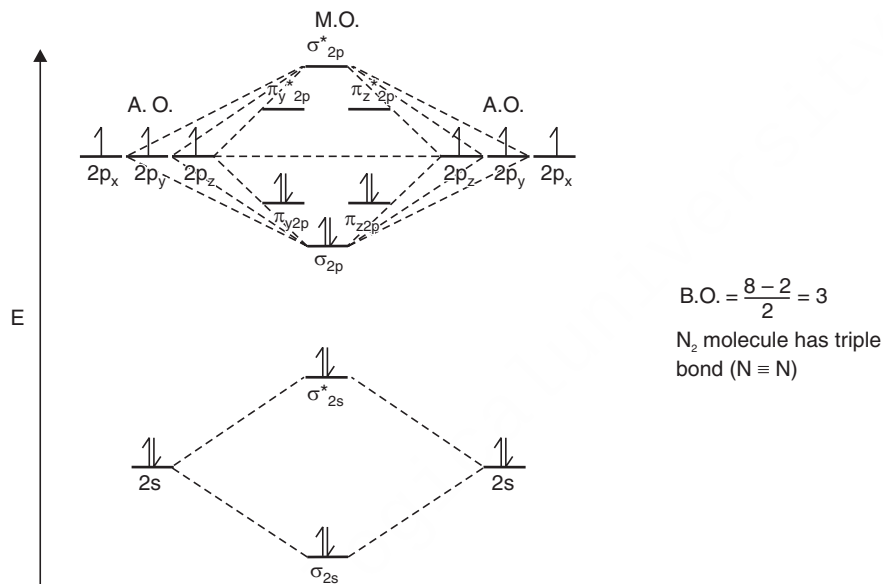
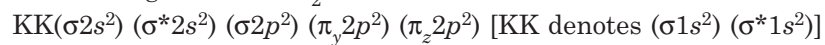


Fig. 2.28 M.O. diagram for N_2 .

- Energy level diagrams of other molecules:

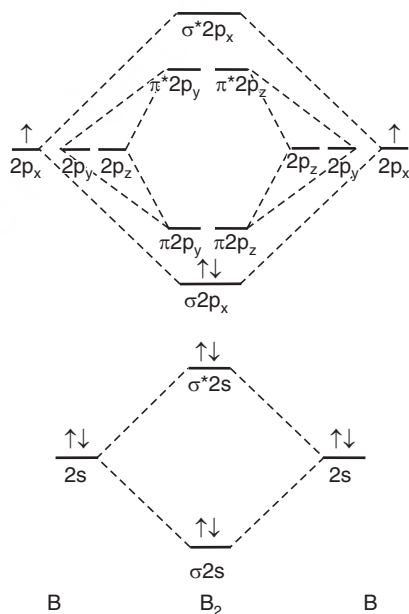


Fig. 2.29 Molecular orbital diagram of B_2 (without s-p interaction).

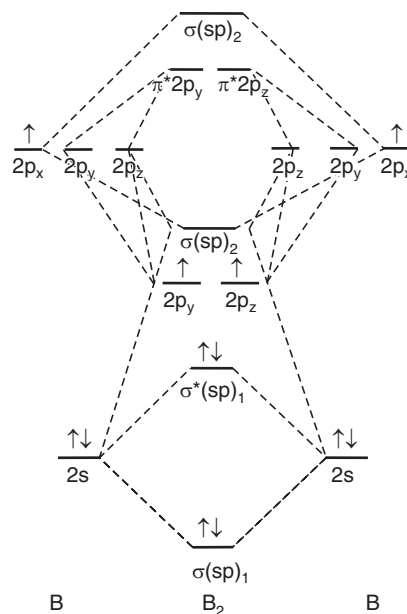
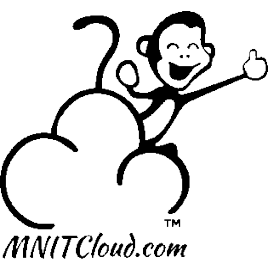


Fig. 2.30 Molecular orbital diagram of B_2 (with s-p interaction).



- Electronic configuration of B_2 molecule:

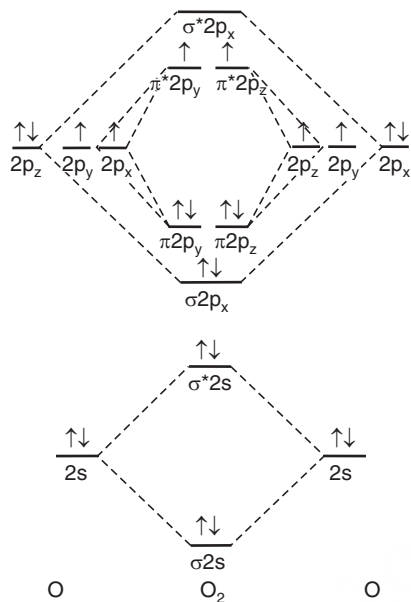
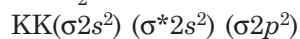


Fig. 2.31 Molecular orbital diagram of O_2 (without s-p interaction).

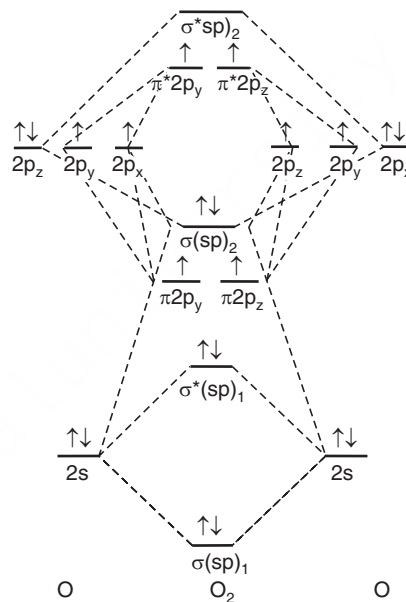


Fig. 2.32 Molecular orbital diagram of O_2 (with s-p interaction).

- Electronic configuration of O_2 molecule:

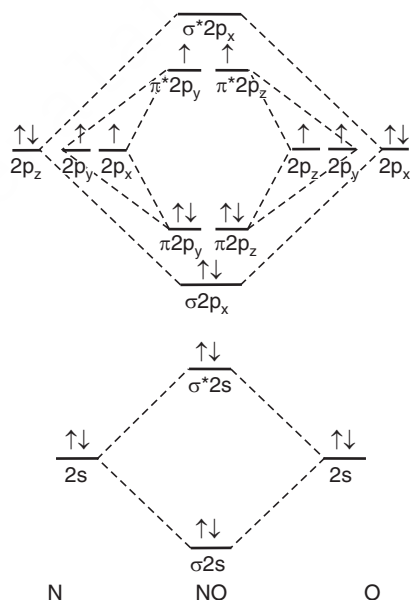
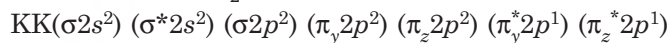


Fig. 2.33 Molecular orbital diagram of NO (without s-p interaction).

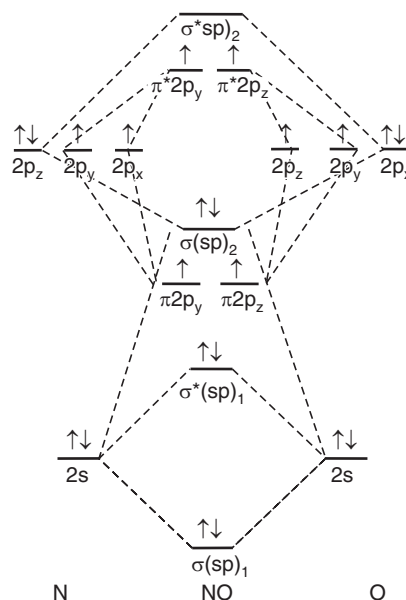


Fig. 2.34 Molecular orbital diagram of NO (with s-p interaction).

